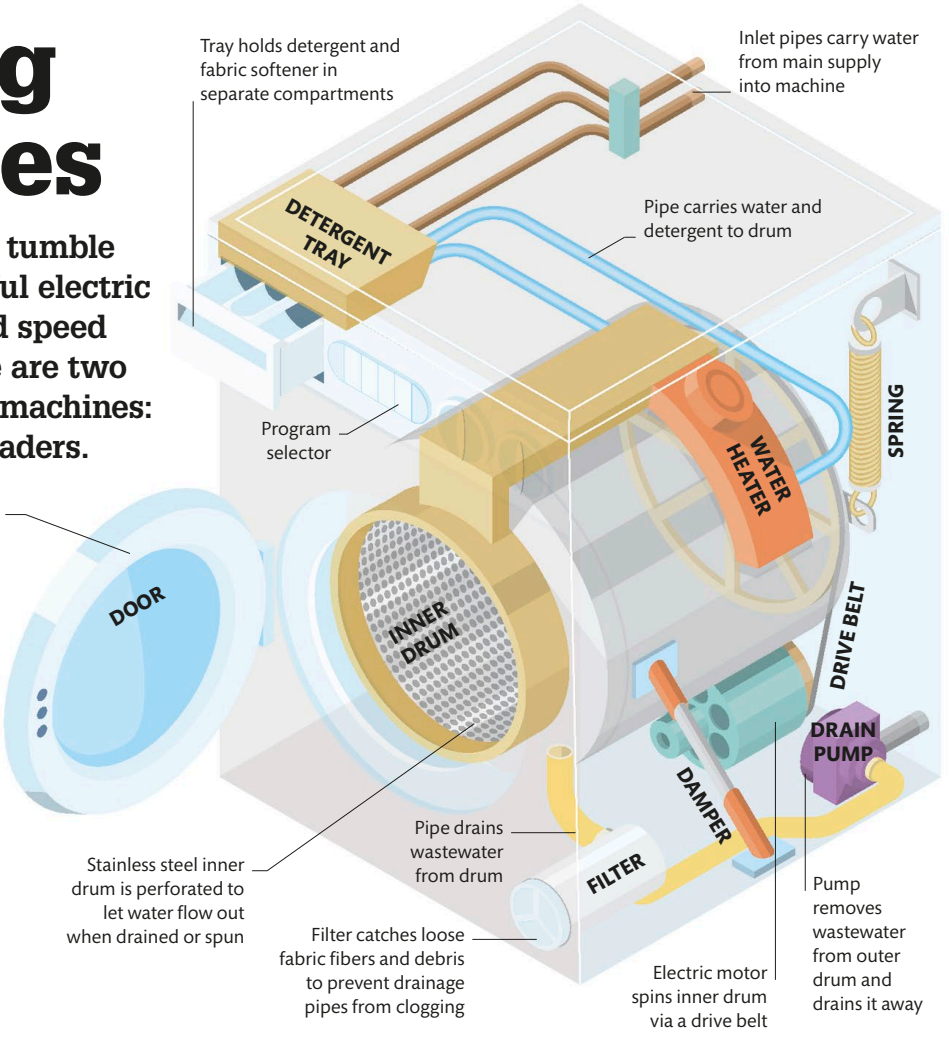
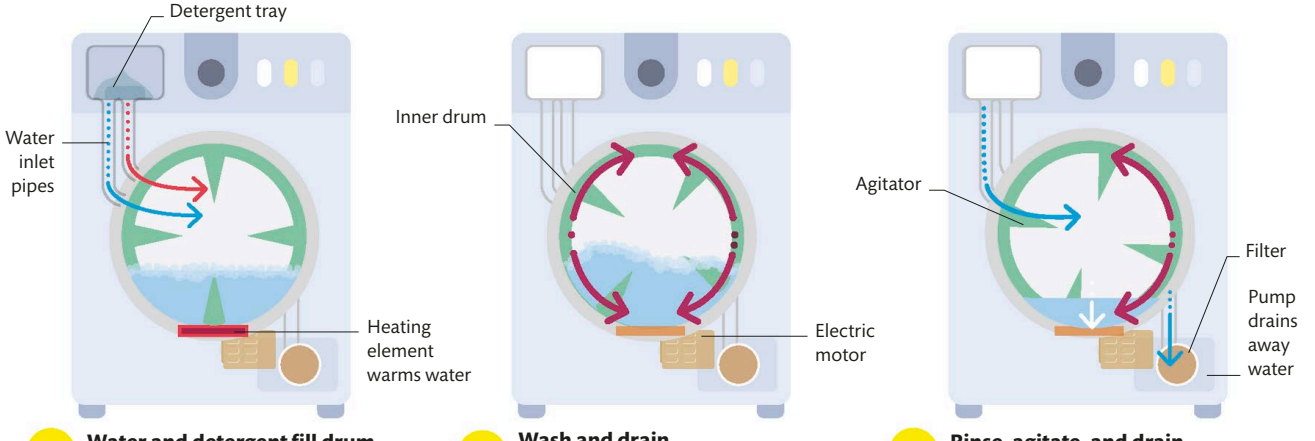
Washing machines

Washing machines and tumble dryers both use powerful electric motors to automate and speed up manual tasks. There are two main types of washing machines: front loaders and top loaders.

Front loaders

An outer drum is held in place inside the washing machine by springs and shock-absorbing dampers. Within it, an inner drum is spun by a motor, either turning slowly to churn water, detergent, and clothes during a wash cycle, or spinning fast to remove water. A program governs the temperature of the water, duration of wash, and the rinsing and spin cycles.





1 Water and detergent fill drum
Water enters the machine and passes through a detergent tray to wash detergent into the drum. Machines may be filled with hot and cold water or with cold water only.

2 Wash and drain
The wash cycle starts once the desired water quantity and temperature are reached. A motor turns the inner drum back and forth through the water-detergent mix.

3 Rinse, agitate, and drain
Wash water is drained away, and cold water fills the machine. Agitators in the inner drum help remove loosened dirt and any detergent remaining on clothes.